Finite PPT Certificates, Projective Geometry, and Secret-Key Boundaries

A Ram Calculus Audit of Manuscript M553

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Abstract

We audit the pinned manuscript *Entanglement with zero distillable secret key in local dimension ten*, separating its finite completely positive and positive-partial-transpose algebra, its degree-twenty projective certificate, and its all-copy operational assertion. An independently reconstructed exact-integer checker reproduces the three recorded modular certificates from the displayed pencil; this is not a replay of the unavailable source program or an independent validation of the headline theorem. We prove a source-specific determinant identity that converts kernel-test errors into squared maximal-minor residuals, compute the composite Choi normalization through a six-dimensional compression, and derive a quantitative full-spark entanglement witness with explicit direction, Gram, and state-error budgets. We also prove that isolated complex rank-loss directions of this real symmetric-Gram pencil must be real, yielding a positive trace-form interpretation of its finite quotient. Using a retained-mass refinement developed in this report, we improve the conditional unhalved secret-bit gap from $1/5$ to $(3-\sqrt{3})/6$ and give an imbalance-sensitive fidelity bound. The actual private Ram passages supply metric-preserving moment coordinates and residual accounting, not a new quantum operational principle. No numerical full-spark margin, protocol optimization, or historical novelty is claimed. Contribution assessment: C3/5, proved residual-to-witness and retained-mass refinements strengthen finite entanglement certification and the conditional secrecy gap while a numerical witness margin and independent validation of the headline construction remain unmet.

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1 Pinned Problem and Exact Scope

1.1 Source and confidentiality

The catalogue row is M553, family 272, at commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The source is dated September 27, 2026. Its local directory, relative to the study root, is

`_sources/openai-math/preprints/Entanglement-with-zero-distillable-secret-key-in-local-dimension-ten-September-27-2026`.

The study root is `C:/Users/mural/OneDrive/Documents/OpenAI-722-math`. In source locators below, filenames in `sections/` and `appendices/` are relative to `build/source/` in that directory. All ten substantive sections and both mathematical appendices were read, including the separate projection and combinatorial construction. The source main file, README, bibliography, and recorded certificate were also inspected. Source hashes and the exact reading scope are retained locally in the author audit; no private corpus text was sent to a public search engine. The catalogue applicability score L5 is not our contribution score. This report follows `contribution-policy.json`.

1.2 Spaces, normalization, and quantifiers

All quantum spaces are finite-dimensional complex Hilbert spaces with fixed orthonormal bases. Write $\mathrm{M}_d = \mathrm{M}_d(\mathbb{C})$, $P_x = xx^*$ for any vector, and $T_d(A) = A^{\mathsf{T}}$ for complex-linear transpose. The adjoint of a map is its Hilbert–Schmidt adjoint, not its entrywise transpose. A map $F : \mathrm{M}_a \to \mathrm{M}_b$ is PPT here if both F and $T_b F$ are completely positive (CP); trace preservation is not included. The Choi convention is input-first and unnormalized:

$$J(F) = \sum_{i,j=1}^{a} E_{ij} \otimes F(E_{ij}), \qquad \Omega_a = \sum_{i=1}^{a} e_i \otimes e_i, \qquad \|\Omega_a\|^2 = a. \tag{1.1}$$

A state is positive of trace one; separability means a finite positive sum of product matrices, equivalently a convex combination of pure product states after normalization. All trace norms below are unhalved.

The proposed dimension-ten state is $\rho = Z/\lambda$, where $Z = J(\Phi_2\Phi_1)$ and $\lambda = \operatorname{tr} Z$. Its advertised properties are entanglement, product-free range, and zero distillable secret key in the source's specified protocol model. The strong operational assertion is

$$\forall n \geq 1 \ \forall \mathcal{P} \ \forall \sigma_{E'} : \quad \left\| \tau^{\mathcal{P},n} - \tfrac{1}{2} \sum_{i=0}^{1} |i,i\rangle\langle i,i| \otimes \sigma_{E'} \right\|_1 \geq \tfrac{1}{5}. \tag{1.2}$$

Here $\mathcal{P}$ must be admitted and completed on $\rho^{\otimes n}$, E' must retain a purification and the complete public transcript, and σ is any normalized comparison law on that output space. The quantifiers are not over a finite set of measured protocols.

There is no spatial PDE boundary in this problem. Its boundaries are instead the projective chart $x_0 \neq 0$, the singular strata of a matrix pencil, the boundary of the positive cone, and the allowed protocol class. The proposed local maps act on all complex matrices. Positivity assertions quantify over every input vector, not just real vectors or sampled states. Projective geometry is over $\mathbb{C}$ until a realness conclusion is proved.

For key strings of length ℓ_n, the rate is $\liminf_n \ell_n/n$, with error tending to zero against a purifying Eve. The supremum of these rates defines the source's K_D. Finite protocols may have round counts increasing with n; the specified countable protocols additionally require compatible complete-history instruments. An abort probability, public postselection, and a trace-norm closure are three different operations. None is silently incorporated into the word "protocol."

2 Stepwise Dependency Audit

The following audit distinguishes a verified finite identity, an inherited deduction whose premises are exposed, and a claim not independently certified here. Reading a proof is not equivalent to independently verifying the theorem as a whole.

Source step	Dependencies and audit disposition
`foundations.tex`, PPT stability, Choi transfer	Kraus formulas, input/output transpose equivalence, and the identity $J(GF) = (F^\sharp \otimes G)(\Omega_b \Omega_b^*)$ are finite identities; $F^\sharp = T_a F^\dagger T_b$. They do not establish nonseparability or secret-key impossibility.

Source step	Dependencies and audit disposition
coherence.tex, lem: coherence	The represented class requires two local CP and input-copositive maps. Support normalization of two pulled-back effects, a spectral split, and two positive-block estimates give the four-effect bound. This is much stronger input than the state's PPT property alone.
transcripts.tex, def:protocol, lem: transcript	Finite transcript factorization is a local-instrument rectangle argument. The countable case additionally uses standard Borel spaces, product probes, conditional independence of full local tapes, matrix-valued measures, and reference-compatible terminal readouts. No finite algebra checker verifies this measure-theoretic assertion.
secret-key.tex, thm:zero-key	Common local effect factors plus the four-effect inequality imply a fidelity sandwich and the 1/5 unhalved gap. Tensor closure and first-bit contraction then yield the all-copy assertion, conditional on those operational hypotheses.
geometry.tex, prop: tensor-criterion	Symmetric and exterior squares, the conjugate exterior dual, twenty singular directions, and independence of every ten quadratic evaluations exclude products from the range of Z. The last independence assertion is arithmetic/projective, not a consequence of a Gram factorization.
certificate.tex, lem:pencil-ppt	The displayed four integer matrices are checked exactly. All block symmetries, the Choi partial transpose, and $M_0^T M_0 = 36I$ pass.
certificate.tex, prop:twenty-directions	Fifteen maximal minors, invertibility of the degree-four coefficient matrix, nonzero quotient, irreducible formal multiplication, and Frobenius cycle types are all required. The three recorded modular outputs are reconstructed exactly; the logical deductions are itemized in Section 4. No complex root list was numerically fitted.
applications.tex	Entanglement and zero key meet only after two independent branches of the proof. The 21-dimensional flag channel inherits nonseparability from an entangled local corner; its full Choi range is not product-free.
projection.tex	A real vector Gram matrix and commuting projection blocks prove PPT. Zero crossed diagonals constrain supports in any complex separable decomposition. The support upper bound must contradict a separate Gram lower bound; PPT alone does not do so.
combinatorial.tex	Restricted intersections, threefold fiber amplification, six parity contexts, and tensor assembly establish the advertised support bound. Only the small binomial/power inequalities are reconstructed here; the enormous global objects are not instantiated.
Both appendices	All mathematical proofs were read. The referenced verification programs and supplementary JSON outputs are absent locally; only verification/certificate-output.txt is available. Our checker is new audit code, not the absent source listing.

3 The Integer Pencil and Its Physical Gram Certificate

The source data, with zero-based indices, are

$$M_0 = \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad M_1 = \begin{pmatrix} -12 & 0 & 0 & 0 \\ 0 & -6 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 12 \\ -6 & 0 & 6 & 6 \\ -6 & -6 & 6 & 0 \end{pmatrix},$$
$$M_2 = \begin{pmatrix} 0 & 6 & -2 & 0 \\ 6 & 6 & 0 & 2 \\ -2 & 0 & 10 & 0 \\ 0 & 2 & 0 & 24 \\ 0 & 0 & 0 & 6 \\ 0 & -6 & 6 & -6 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 0 & 0 & -4 & -3 \\ 0 & -2 & -3 & -4 \\ -4 & -3 & 11 & 6 \\ -3 & -4 & 6 & 25 \\ 6 & 6 & 0 & 0 \\ 0 & -6 & 6 & 6 \end{pmatrix}. \tag{3.1}$$

Put $M(x) = \sum_{i=0}^{3} x_i M_i$ and define $L(E_{ij}) = M_i^T M_j$. Exact integer multiplication in the checker verifies

$$M_i^T M_j = M_j^T M_i \quad (0 \le i, j < 4), \qquad M_0^T M_0 = 36 I_4. \tag{3.2}$$

Proposition 3.1 (Finite certificate, with complex conventions)**.** *The displayed $L : \mathrm{M}_4 \to \mathrm{M}_4$ is CP, satisfies $T_4 L = L$, and is PPT. For every $x \in \mathbb{C}^4$,*

$$L(P_x) = M(x)^T \overline{M(x)} = M(x)^* M(x), \qquad \ker L(P_x) = \ker M(x). \tag{3.3}$$

Proof. Let Q_r have as its ith column the transpose of row r of M_i. Then $\sum_r Q_r E_{ij} Q_r^* = M_i^T M_j$, so L has a Kraus decomposition. Equivalently $J(L) = B^T B \succeq 0$ with $B = [M_0 \; M_1 \; M_2 \; M_3]$. Each block is symmetric by (3.2), so output partial transpose fixes $J(L)$ and $T_4 L = L$. Expanding P_x gives the first expression in (3.3). Interchange i, j and apply (3.2) to obtain the usual conjugate Gram expression. Its quadratic form at v is $\|M(x)v\|^2$, proving the kernel identity even at singular inputs. □

The equality of the two complex Gram expressions is special to (3.2); it cannot be inferred from reality of the M_i alone. No numerical eigenvalue tolerance is needed. The checker also computes $\operatorname{rank} J(L) = 6$ and directly tests its 16×16 partial transpose. This rank computation is finite rational elimination, not a test of the rank of the later 100×100 composite.

4 Projective Quotient and the Arithmetic Gate

Let $\Delta_\rho(x)$ run over the fifteen 4×4 minors of $M(x)$, with increasing row sets $\rho \subset \{0, \dots, 5\}$. The singular locus in $\mathbb{P}^3(\mathbb{C})$ is their common zero set. In the chart $x_0 = 1$ put $p_\rho(t) = \Delta_\rho(1, t_1, t_2, t_3)$ and $\mathcal{A} = \mathbb{Q}[t_1, t_2, t_3]/(p_\rho)$. For total degree d, order monomials as $(a, b, d-a-b)$, with $a = 0, \dots, d$ and $b = 0, \dots, d-a$. The twenty monomials of degree at most three form a column v, and the fifteen of degree four form w. Define integer coefficient matrices by

$$(p_\rho)_\rho = Dv + Hw, \qquad D \in \mathbb{Z}^{15 \times 20}, \quad H \in \mathbb{Z}^{15 \times 15}. \tag{4.1}$$

The independent checker expands determinants by their 24 permutations, with integer polynomial dictionaries, and reconstructs all coefficients.

4.1 Executed modular evidence

For $C = -H^{-1}D$, define the formal 20×20 matrix N by multiplying a low monomial by t_1 and replacing a degree-four product by the corresponding row of C. Let $f(z) = \det(zI - N)$. The computation performs the same construction modulo each prime, with column-oriented multiplication. Newton identities calculate all 21 characteristic coefficients, and polynomial powering and Euclidean division calculate all twenty degrees $d_p(k) = \deg \gcd(\bar{f}, z^{p^k} - z)$.

p	$\det H \bmod p$	Sufficient gcd degrees	Factor degrees
41	40	$d(k) = 0$ for $k < 20$; $d(20) = 20$	20
131	23	$d(1) = 1$, $d(2) = 3$, $d(17) = 18$	1,2,17
139	9	$d(1) = 1$, $d(19) = 20$	1,19

Every coefficient and every gcd degree agrees with the recorded file, not merely these selected entries. The reconstructed multiplication operators for t_1, t_2, t_3 commute modulo all three primes. That extra check is a diagnostic only: congruences at three primes are not a proof of rational commutation without a height bound.

4.2 What is still logically required

Here is the full algebraic implication chain, with no numerical-root substitute.

1. Nonzero determinant modulo 41 proves that H is rationally invertible. At $x_0 = 0$, the minors are $Hw(t)$; their common vanishing would force $t_1^4 = t_2^4 = t_3^4 = 0$. Thus there are no points at infinity.

2. A common projective zero exists because six bilinear equations $M(x)y = 0$ on $\mathbb{P}^3 \times \mathbb{P}^3$ have a common zero. One proof uses the affine Segre cone: its coordinate ring has the strict prime chain obtained by successively suppressing rows and columns, ending at the vertex maximal ideal with height at least seven. If six linear equations on that cone vanished only at the vertex, Nullstellensatz and Krull's height theorem would force that maximal ideal to have height at most six. This contradiction gives nonzero x, y. With no point at infinity, the affine quotient is nonzero.

3. The relations $w = Cv$ lower total degree, so every monomial is spanned by the twenty low monomials. This proves spanning, not yet independence or unique reduction.

4. The natural surjection $\pi : \mathbb{Q}^{20} \to \mathcal{A}$ intertwines N with multiplication by t_1. Since $\bar{f}$ is irreducible modulo 41 and f is monic with coefficients integral at 41, f is irreducible over $\mathbb{Q}$. To justify the local-integrality step, the minimum p-adic coefficient valuation is additive under polynomial products; monic factors of an integral monic polynomial therefore remain integral.

5. A nonzero proper N-invariant rational subspace would factor its characteristic polynomial. Hence $\ker \pi = 0$ and the twenty monomials are a basis. The minimal polynomial of N is f, so $\mathcal{A} = \mathbb{Q}[t_1] \simeq \mathbb{Q}[z]/(f)$ is a degree-twenty field.

6. Characteristic zero makes f separable. Over $\mathbb{C}$ the quotient is $\mathbb{C}^{20}$; its points are $(\lambda_j, h_2(\lambda_j), h_3(\lambda_j))$, where h_2, h_3 are rational polynomials of degree below twenty. This gives precisely twenty distinct reduced affine points, and the infinity argument gives all projective points.

7. Squarefree reductions at 131 and 139 give Frobenius cycle types $(1, 2, 17)$ and $(1, 19)$. This step invokes the locally integral Frobenius theorem, whose lifting proof is supplied in the source appendix. A transitive group containing a 19-cycle is two-transitive: the stabilizer of its fixed point acts transitively on the other points. The seventeenth power of the first cycle is a transposition. Two-transitivity conjugates it to every transposition, giving S_{20}.

8. The ten degree-at-most-two monomials belong to the quotient basis. Their evaluation matrix has column rank ten, so one ten-row determinant is nonzero. The S_{20} action transports that algebraic determinant to each ten-element subset. Every such determinant is therefore nonzero over $\mathbb{C}$, including against arbitrary complex coefficient vectors.

These arguments clarify exactly what the finite modular replay supports. They use classical projective algebra, local field theory, and linear algebra, not a new Ram irreducibility theorem. We do not characterize the entire source theorem as independently verified on the strength of this chain and one checker; the projective proof remains a separate logical dependency of the finite construction.

5 A Realness and Trace-Metric Reformulation

The special complex Gram identity yields an additional structural test not stated as such in the source's projective certificate.

Theorem 5.1 (Isolated singular directions are real)**.** *Let real matrices M_i satisfy $M_i^T M_j = M_j^T M_i$. If the projective rank-loss set of $M(x) = \sum x_i M_i$ is finite, every one of its points has a real representative. In particular, conditional on the finite degree-twenty quotient conclusion above, all twenty M553 directions have real representatives $(1, t_1, t_2, t_3)$.*

Proof. Write $x = a + ib$, with a, b real. The block identities imply $M(a)^T M(b) = M(b)^T M(a)$. Hence

$$M(x)^* M(x) = M(a)^T M(a) + M(b)^T M(b).$$

If $M(x)y = 0$ for a nonzero complex y, its Gram quadratic form gives $\|M(a)y\|^2 + \|M(b)y\|^2 = 0$. Both terms vanish. Therefore $M(sa + tb)y = 0$ for every $s, t \in \mathbb{C}$. If a, b were real-linearly independent, they would also be complex-linearly independent, giving a whole projective line of singular directions. Finiteness rules this out. Thus x is a complex scalar multiple of a nonzero real vector. If $x_0 \neq 0$, scaling to first coordinate one makes all other coordinates real. □

This proof handles a complex kernel vector and does not incorrectly replace complex rank loss by a real numerical search. It also explains why a non-real isolated root returned by approximate elimination would be a serious inconsistency to investigate.

Corollary 5.2 (Positive quotient trace metric)**.** *Assume the degree-twenty field identification of Section 4. Let $b_\alpha = t^\alpha$, $|\alpha| \leq 3$, and N_α be exact multiplication by b_α. Then*

$$G_{\alpha\beta} = \frac{1}{20} \operatorname{tr}(N_\alpha N_\beta) = \frac{1}{20} \sum_{j=1}^{20} b_\alpha(t^{(j)}) b_\beta(t^{(j)}) \tag{5.1}$$

is a positive definite rational matrix. For complex coefficient vectors c, $c^* G c = \frac{1}{20} \sum_j |\sum_\alpha c_\alpha b_\alpha(t^{(j)})|^2$.

Proof. The multiplication operators diagonalize in the split algebra with entries given by evaluations, proving the trace formula. The preceding theorem makes all evaluations real. The square evaluation matrix on a quotient basis is invertible over $\mathbb{C}$, so the displayed sum of squared absolute values vanishes only for $c = 0$. Rationality follows from the rational multiplication matrices. This is the normalized field trace form of a totally real field, not an assumed positive form for an arbitrary number field. □

This provides an exact metric construction without selecting approximate roots. We have not computed its rational entries or a lower eigenvalue bound. A positive metric on all twenty evaluations does not by itself give the minimum metric on every ten-row subset, which is needed next. For a general degree-twenty field with complex embeddings, the bilinear trace matrix $\mathrm{tr}(N_\alpha N_\beta)$ need not be positive and is not a Hermitian evaluation Gram. Positivity here depends on the preceding realness theorem and the finite reduced quotient, not on the degree or Galois group alone. No rational involution or assertion that complex conjugation commutes with all Galois automorphisms is used.

6 Tensor Construction and Exact Normalization

Use lexicographic pair bases. Define real isometries $U : \mathbb{C}^{10} \to \mathbb{C}^4 \otimes \mathbb{C}^4$ and $V : \mathbb{C}^6 \to \mathbb{C}^4 \otimes \mathbb{C}^4$ by

$$Ue_{ii} = e_i \otimes e_i, \quad Ue_{ij} = (e_i \otimes e_j + e_j \otimes e_i)/\sqrt{2}, \quad Ve_{ij} = (e_i \otimes e_j - e_j \otimes e_i)/\sqrt{2}.$$

Thus $\widehat{x} = U^*(x \otimes x)$ has diagonal coordinates x_i^2 and off-diagonal coordinates $\sqrt{2}x_i x_j$, and $\|\widehat{x}\| = \|x\|^2$. On exterior coordinates $(01, 02, 03, 12, 13, 23)$ let

$$(Ke_{01}, Ke_{02}, Ke_{03}, Ke_{12}, Ke_{13}, Ke_{23}) = (e_{23}, -e_{13}, e_{12}, e_{03}, -e_{02}, e_{01}).$$

Then $K = K^* = K^{-1}$. Define rectangular maps

$$S(A) = V^*(L \otimes L)(UAU^*)V, \qquad R(A) = KS(A)^T K, \quad S, R : \mathrm{M}_{10} \to \mathrm{M}_6\,. \tag{6.1}$$

Both are PPT: CP embeddings and compressions preserve this property, as do tensor products and an output transpose of a PPT map. If $E : \mathbb{C}^6 \hookrightarrow \mathbb{C}^{10}$ is the first-coordinate inclusion,

$$\Phi_1(A) = ES(A^T)E^*, \qquad \Phi_2(B) = R^\dagger(E^*BE).$$

Their composition is $R^\dagger S T_{10}$, and $Z = J(R^\dagger S T_{10})$ is positive. For Hermitian a, b,

$$\mathrm{tr}[(a \otimes b)Z] = \mathrm{tr}[R(b)S(a)]. \tag{6.2}$$

Indeed the input-first Choi pairing applies F to a^T; this cancels the displayed input transpose, after which the adjoint identity applies.

For $B_x = L(P_x)$, $S(P_{\widehat{x}}) = \bigwedge^2 B_x$. If $\mathrm{rank}\, B_x < 4$, its range is $\bigwedge^2 \mathrm{ran}\, B_x$, whose conjugate complementary image under K is orthogonal to it: for two exterior vectors w, u in that space,

$$\langle K\overline{w}, u\rangle = [w \wedge u]_{0123} = 0.$$

The conjugation is essential for complex subspaces. Equation (6.2) then places $\widehat{x} \otimes \widehat{x}$ in $\ker Z$, using positivity of Z. At $x = e_0$, the two exterior outputs are $36^2 I_6$, so

$$(e_{00} \otimes e_{00})^* Z(e_{00} \otimes e_{00}) = 6 \cdot 36^4 = 10\,077\,696 > 0. \tag{6.3}$$

If every ten $\widehat{x}_j$ span, a nonzero product in $\mathrm{ran}\, Z$ would give two nonzero quadratics whose zero sets together cover twenty points; each can cover at most nine. This is the source's qualitative exclusion.

6.1 A six-dimensional normalization computation

The Choi trace is obtained without constructing the full composite:

$$\lambda = \operatorname{tr} Z = \operatorname{tr}[R(I_{10})S(I_{10})] = 130\,208\,350\,782. \tag{6.4}$$

For reproducibility, set $A_{ij} = M_i^T M_j$, $B = \sum_i A_{ii}$, and define a 16×16 tensor-indexed array

$$T_{ab,cd} = \tfrac{1}{2}\left(B_{ac}B_{bd} + \sum_{i,j=0}^{3} (A_{ij})_{ac}(A_{ji})_{bd} \right). \tag{6.5}$$

Since $UU^* = (I + \mathrm{Swap})/2$, this is exactly $(L \otimes L)(UU^*)$. The six-dimensional array is

$$D_{ab,cd} = \tfrac{1}{2}(T_{ab,cd} - T_{ab,dc} - T_{ba,cd} + T_{ba,dc}), \qquad a < b,\ c < d.$$

It equals $S(I_{10})$, and $R(I_{10}) = KD^TK$. Exact rational arithmetic evaluates $\operatorname{tr}(KD^TKD)$ to (6.4). This is an executed normalization computation; it is not an independently measured entanglement or key rate. All intermediate entries of D are recorded in the check JSON.

7 From Minor Residuals to a Robust Entanglement Witness

This section supplies the main local extension. It replaces an exact-zero kernel premise by an explicit positive-tolerance entanglement test. It applies to the same finite maps and has no protocol conclusion.

Theorem 7.1 (Exact minor-to-kernel residual identity)**.** *For the M553 tensor construction and every $x \in \mathbb{C}^4$,*

$$(\widehat{x} \otimes \widehat{x})^* Z (\widehat{x} \otimes \widehat{x}) = 6 \det L(P_x) = 6 \sum_{|\rho|=4} |\Delta_\rho(x)|^2. \tag{7.1}$$

Proof. Let $C = \bigwedge^2 B_x$, where $B_x = L(P_x) \succeq 0$. Preservation of the exterior wedge pairing under B_x gives $C^T K C = (\det B_x)K$: both sides express the coefficient of the top exterior power after applying B_x to four vectors. This identity holds also when B_x is singular, either directly by multilinearity or as a polynomial identity. Multiplication by K gives $KC^TKC = (\det B_x)I_6$. Equations (6.1) and (6.2) now yield the first equality by taking the trace. For the second, use $B_x = M(x)^*M(x)$ and Cauchy–Binet, with the conjugate minors from $M(x)^*$. No real-input restriction or singular inverse appears. □

For any chosen nonzero directions $x_1, \ldots, x_N$ define unit vectors $v_j = \widehat{x}_j / \|x_j\|^2 \in \mathbb{C}^{10}$ and the positive test matrix

$$F = \sum_{j=1}^{N} (v_j \otimes v_j)(v_j \otimes v_j)^*. \tag{7.2}$$

The next theorem is stated in general dimension d and then applied with $d = 10, N = 20$.

Theorem 7.2 (Quantitative full-spark product exclusion)**.** *Let unit vectors $v_1, \ldots, v_N \in \mathbb{C}^d$ satisfy $N \geq 2d - 1$ and suppose every d of them span. Set*

$$s = \min_{|I|=d} \lambda_{\min}\left(\sum_{j \in I} v_j v_j^* \right) > 0, \qquad \mu = (N - 2d + 2)\frac{s^2}{d^2}. \tag{7.3}$$

Then $W = F - \mu I_{d^2}$ is nonnegative on every unit product vector, and $\operatorname{tr}(F\sigma) \geq \mu$ for every separable density matrix σ. Every density matrix τ with $\epsilon = \operatorname{tr}(F\tau) < \mu$ is entangled and satisfies

$$\inf_{\sigma \text{ separable}} \|\tau - \sigma\|_1 \geq \frac{2(\mu - \epsilon)}{\|F\|_\infty} \geq \frac{2(\mu - \epsilon)}{N}. \tag{7.4}$$

Proof. The minimum in (7.3) is positive because there are finitely many subsets and each frame operator is positive definite. Fix a unit u. Fewer than d indices can have $|v_j^* u|^2 < s/d$; otherwise the sum on those d indices would be less than s, contradicting its frame bound. Thus at least $N - d + 1$ indices have squared overlap at least s/d. For a second unit vector w, its corresponding index set intersects the first in at least $N - 2d + 2$ indices. Consequently

$$\langle u \otimes w, F(u \otimes w)\rangle = \sum_j |v_j^* u|^2 |v_j^* w|^2 \geq \mu.$$

Convexity gives the separable bound and hence the entanglement test. For the distance bound, $A = F/\|F\|_\infty$ is an effect, $0 \leq A \leq I$. For a trace-zero Hermitian H, spectral positive/negative decomposition gives $|\operatorname{tr}(AH)| \leq \|H\|_1 /2$. Apply this with $H = \sigma - \tau$. Finally F is a sum of N unit rank-one projections, so $\|F\|_\infty \leq N$. □

Corollary 7.3 (M553 residual threshold)**.** *For twenty nonzero candidate directions satisfying the ten-point frame condition with certified lower bound $s \geq s_0 > 0$, the exact M553 state is entangled whenever*

$$\frac{6}{130\,208\,350\,782} \sum_{j=1}^{20} \frac{\sum_{|\rho|=4} |\Delta_\rho(x_j)|^2}{\|x_j\|^8} < \frac{s_0^2}{50}. \tag{7.5}$$

The directions need not be exact algebraic roots or exact rank-loss directions. Its unhalved distance to separable states is at least $(s_0^2/50 - \epsilon)/10$, with ϵ the left side.

Proof. Theorem 7.1, normalization (6.4), and (7.2) give exactly the displayed value of $\operatorname{tr}(F\rho)$. Use Theorem 7.2 with $d = 10, N = 20$ and $N - 2d + 2 = 2$. Replacing s by a proved lower bound can only weaken the valid witness. □

For the exact source directions, the residual is zero and the arithmetic gate implies $s > 0$. This recovers its qualitative conclusion. The new use is a finite certified nonzero-residual route to entanglement, avoiding an unsupported inference from small residuals to exact ideal membership. There are $\binom{20}{10} = 184\,756$ frame subsets: this is finite but not negligible. We have not evaluated their minimum, and the report therefore supplies no numerical robustness radius for the actual state. The threshold is an explicit theorem and reproducible verification contract, not an executed experimental performance claim.

7.1 Direction, metric, and state errors

Proposition 7.4 (Complete finite error budget)**.** *Suppose unit $\widetilde{v}_j$ approximate the intended v_j and*

$$\left\| v_j v_j^* - \widetilde{v}_j \widetilde{v}_j^* \right\|_\infty \leq \delta \quad \text{for all } j.$$

If every approximate d-subset frame has a certified lower bound $\widetilde{s}$, then $s \geq \widetilde{s} - d\delta$. Moreover, for any density τ,

$$\operatorname{tr}(F\tau) \leq \operatorname{tr}(\widetilde{F}\tau) + 2N\delta. \tag{7.6}$$

If another normalized state τ' obeys $\|\tau' - \tau\|_1 \leq r$, the right side can be increased by $\|F\|_\infty r/2$ to bound $\operatorname{tr}(F\tau')$.

Proof. Sum the projector errors over a d-subset and apply the Rayleigh variational characterization of the least eigenvalue. For projectors P, Q of norm one, $P \otimes P - Q \otimes Q = (P-Q) \otimes P + Q \otimes (P-Q)$, so its norm is at most 2δ. Sum over N and use positivity and unit trace of τ. The last assertion follows from the trace-zero effect bound in the proof of Theorem 7.2. □

The criterion is useful only when $\widetilde{s} - d\delta > 0$ and the total residual budget is strictly below its resulting μ. Intervals containing zero do not certify zero minors. Approximate PSD matrices require their own PSD and normalization correction before they can be called density matrices in this proposition.

One exact direction can be discarded while preserving qualitative exclusion: nineteen directions still give $N = 2d-1$ and the same ten-point independence premise, with $\mu = s^2/100$ for the retained frame. Two discarded directions would leave no positive bound from this argument. This is an erasure statement about the finite witness, not permission to erase Eve's transcript.

8 Specific Ram Correspondences and Their Limits

8.1 Passages actually read

The following are private local references, not public literature and not substitutes for the source proof.

1. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex`. The entire chapter file was read, especially `v21c01:physical-gram-definition`, `v21c01:metric-symmetry`, `v21c01:residual-decomposition`, `v21c01:residual-identity`, and `v21c01:gram-equivalence`. Its concrete lesson is to transport the physical metric and account for omitted components; a finite Gram matrix is not a continuum or asymptotic theorem.

2. `C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`. Read "Physical spaces precede matrices," "What a Ram representation means in this report," equation `eq:moments`, "Ram moment coordinates that retain the complex," theorem `thm:transfer`, and "A concrete polynomial Ram construction." The report explicitly identifies invertible finite coordinate transfer as preservation of a classical guarantee, not a new underlying theorem.

3. `C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/res-math/TR-item-047-astra.tex`. Read "Algebraic and Moment Reformulation," "Ram Analysis and Synthesis Panels," "Global Optimality Sandwich," "Making the Moment and Identity Tests Exact," and "Residual-Aware SOS Lower Bounds," including `eq:gap`, `sec:soserror047`, and `eq:gramerror047`. The relevant distinction is exact coefficient identity versus a certified residual charged to the final inequality.

8.2 A concrete ten-dimensional moment bank

Let $\mathcal{P}_2$ be the complex polynomials in $t = (t_1, t_2, t_3)$ of total degree at most two, with the ten monomials $m_\alpha = t^\alpha$ in the source's order. Define the moments on the real unit cube by

$$\ell_\alpha(p) = \int_{[0,1]^3} t^\alpha p(t)\, dt, \qquad H_{\alpha\beta} = \prod_{h=1}^{3} \frac{1}{\alpha_h + \beta_h + 1}, \qquad m = Hc. \tag{8.1}$$

The cube is merely a coordinate-design panel; it is not claimed to contain the source's projective points. H is rational positive definite: $c^*Hc = \int |\sum c_\alpha t^\alpha|^2 > 0$ for $c \neq 0$. Thus $R = H^{-1}$ reconstructs $c = Rm$ exactly. This is precisely the finite polynomial Ram analysis/synthesis pattern in the cited passages.

The physical norm for a homogeneous quadratic is not this cube norm. Write $p_u(x) = u^*\widehat{x}$ in the ten homogeneous monomials; the coefficient vector c then satisfies

$$\|u\|^2 = c^*Gc, \qquad G = \operatorname{diag}(g_\alpha), \quad g_\alpha = \begin{cases} 1 & \text{square monomial,} \\ 1/2 & \text{mixed monomial.} \end{cases} \tag{8.2}$$

The four squares are $x_0^2, x_1^2, x_2^2, x_3^2$; the six mixed terms inherit the $\sqrt{2}$ Veronese normalization. In moment coordinates, the physical metric is $G_R = R^*GR$, not I and not H.

For a set of normalized directional evaluations, let E_I have rows $m(x_j)^T / \|x_j\|^2$, with homogeneous monomials before dehomogenization. The exact subset frame test is

$$E_I^*E_I \succeq s_0 G \quad \Longleftrightarrow \quad R^*E_I^*E_IR \succeq s_0 G_R. \tag{8.3}$$

The map from u to c in (8.2) is conjugate-linear, but is bijective and norm-preserving with that metric; hence the quantifier over all coefficient vectors is unchanged. This accounts for both the physical norm and complex conjugation in the evaluation test.

Proposition 8.1 (Residual-aware moment certificate)**.** *Suppose a Hermitian computed matrix $\widehat{A}_I$ satisfies $\widehat{A}_I \succeq aG_R$ and the exact evaluation Gram obeys*

$$-\eta G_R \preceq R^*E_I^*E_IR - \widehat{A}_I \preceq \eta G_R.$$

Then the exact frame bound for this subset is at least $a - \eta$. If the same bounds hold for every ten-element subset and $a > \eta$, one may set $s_0 = a - \eta$ in Corollary 7.3.

Proof. Add the first inequality to the lower residual inequality and apply the congruence equivalence (8.3). Equivalently, whiten by $G_R^{-1/2}$ and subtract the operator-norm error from a lower eigenvalue bound. This works for every coefficient vector, not just the computed approximate minimizer. □

This is the concrete correspondence with the corpus's physical metric and inexact Gram/identity budget. Unlike a generic statement that "Ram moments apply," it specifies the ten functionals, rational analysis matrix, physical metric, transformed inequality, and failure threshold. It does not show faster arithmetic or better conditioning; those would require measured comparisons. No new integral transform is needed for (7.1); that identity is classical exterior algebra and Cauchy–Binet in a source-specific application.

9 Operational Audit Without a Finite-to-Asymptotic Leap

9.1 Representations and tensor powers

The source class $\mathscr{C}$ consists of normalized states

$$\omega = (\mathcal{L} \otimes \mathcal{M})(\chi\chi^*), \qquad \mathcal{L}, \mathcal{L}T_r, \mathcal{M}, \mathcal{M}T_s \text{ CP},$$

where $\chi \in \mathbb{C}^r \otimes \mathbb{C}^s$ may be unnormalized and the maps may be rectangular and non-trace-preserving. Tensoring these maps and regrouping local factors proves finite tensor closure. The rectangular Choi identity gives

$$\rho = (\Phi_1^\sharp \otimes \Phi_2)\left(\Omega_{10}\Omega_{10}^*/\lambda\right), \qquad \left\|\Omega_{10}/\sqrt{\lambda}\right\|^2 = 10/\lambda.$$

No extra factor of ten is inserted. Membership in $\mathscr{C}$ is an algebraic representation property; the non-normalized maps are not asserted to be physically implementable deterministic channels.

9.2 Four-effect dependency

For positive X_i, Y_j, put $D_{ij} = X_i \otimes Y_j$ and $p_{ij} = \operatorname{tr}(D_{ij}\omega)$. The source proves

$$\left\| \sqrt{D_{00}}\omega\sqrt{D_{11}} \right\|_1 \le \sqrt{p_{00}p_{01}} + \sqrt{p_{10}p_{11}} + 2\sqrt{p_{01}p_{10}}. \tag{9.1}$$

Its key dependencies can be checked locally as follows. Pull effects back to $A_i = \mathcal{L}^\dagger(X_i)$ and $B_j = \mathcal{M}^\dagger(Y_j)$. Let $H = A_0 + A_1$ and define S as $H^{-1/2}$ on its range and identity on its kernel. This is invertible, and replacing $\mathcal{L}$ by $\mathcal{L}\operatorname{Ad}_S$ and χ by $(S^{-1} \otimes I)\chi$ preserves the output and copositivity. The new A_0, A_1 sum to a projection and commute.

Split $\chi = v + w$ by the common spectral projection where $A_0 \ge A_1$. All branch quadratic weights split additively. A positive two-by-two block matrix with off-diagonal Z satisfies $\|Z\|_1 \le \sqrt{\operatorname{tr} H_0 \operatorname{tr} H_1}$, by factoring it into block columns and applying Cauchy–Schwarz. Applied to $C = \mathcal{L} \otimes \mathcal{M}$, it gives

$$\left\| \sqrt{D_{00}}C(uz^*)\sqrt{D_{11}} \right\|_1 \le \sqrt{p_{00}(u)p_{11}(z)}.$$

Applied to the CP map $CT_{rs} = (\mathcal{L}T_r) \otimes (\mathcal{M}T_s)$ on $(\bar{z}, \bar{u})$, it gives the same off-diagonal output with the diagonal weights reversed. The spectral inequalities $p_{11}(v) \le p_{01}(v)$ and $p_{00}(w) \le p_{10}(w)$ bound the two diagonal expansion terms and the two mixed terms, yielding (9.1). Ordinary PPT of ω alone does not provide this representing-map argument.

9.3 Exactly what transcript factorization requires

For a finite, finite-record local protocol, at a fixed complete public transcript t, private histories sum independently inside each party. The four complete branch maps are $\mathcal{A}_i^t \otimes \mathcal{B}_j^t$ with common local pairs. With the canonical purification of $R = \omega^{\otimes n}$, the Eve blocks are therefore

$$\tau_{ij}(t) = \left(\sqrt{R}[X_i(t) \otimes Y_j(t)]\sqrt{R}\right)^T. \tag{9.2}$$

For a continuous transcript this is a density with respect to one common finite measure, not a conditional state at a positive-probability singleton. Separate factorizations for the four branches would be insufficient for (9.1).

The source's countable extension requires the following retained data: an input-independent public law; conditionally independent local private auxiliaries; standard Borel outcomes; reference-compatible prefix laws; public speaker and stop choices in the record; and a supplied local CP memory process for every terminal quantum readout. On product probes, the two local tapes are conditionally independent given finite public prefixes. Increasing-sigma-field convergence of conditional expectations extends the scalar independence to the full public record. An informationally complete finite reference POVM converts it to product matrix densities. A fixed filter $\sqrt{ab}(\sqrt{R})^T$ on the untouched probe references then recovers the actual purification. Missing outputs may be locally defaulted on the probes; positivity makes their filtered contribution zero when the actual protocol completes almost surely.

These are genuine additional premises and measure-theoretic steps. There is no assertion here that every vaguely specified infinite quantum evolution supplies a terminal readout, or that every element of a topological LOCC closure carries a recoverable full transcript. The source deliberately treats those issues separately. Our finite checker neither tests nor establishes them.

9.4 Conditional gap derivation and stability boundary

Assume the common factors (9.2) and (9.1) for the actual input. Write P_{ij} for the integrated branch probabilities, $e = P_{01} + P_{10}$, and

$$\mathfrak{f} = \int \left\| \sqrt{\tau_{00}(t)}\sqrt{\tau_{11}(t)} \right\|_1 d\mu(t).$$

Polar decomposition gives pointwise equality of this fidelity with the left side of (9.1). Integration and Cauchy–Schwarz give $\mathfrak{f} \le \sqrt{e(1-e)} + e$. If η is the trace-norm error against an ideal bit, the unequal-bit blocks contribute e, and the sum of errors on equal-bit blocks is at least e by trace conservation. Thus $e \le \eta/2$. The difference of the two actual equal-bit blocks has integrated norm at most $\eta - e$. The positive-matrix inequality

$$\left\| \sqrt{A}\sqrt{B} \right\|_1 \ge \tfrac{1}{2}(\operatorname{tr} A + \operatorname{tr} B - \|A - B\|_1)$$

then gives $\mathfrak{f} \ge (1-\eta)/2$. At $\eta < 1/5$ the upper bound is less than $2/5$ and the lower bound greater than $2/5$. This proves the gap under these premises. Keeping the first bit of a longer uniform key is a channel, so contraction rules out a secure sequence with even one output bit infinitely often. The tensor and transcript arguments, not the numerical constant, carry the all-copy and all-protocol force of this deduction.

9.5 A retained-mass improvement

The retained-mass refinement developed here preserves information lost when the four terms in the source proof are enlarged separately. We give the full argument, including the measurable-envelope step. It changes a quantitative bound under the existing hypotheses, not the protocol class or the finite construction.

Theorem 9.1 (Retained-mass fidelity bound)**.** *Assume a normalized represented input and common positive local effect densities for the four final bit branches, as in (9.2). Let P_{ij} and $\mathfrak{f}$ be as above, and put*

$$T = P_{00} + P_{11}, \qquad m = 2\sqrt{P_{01}P_{10}}, \qquad \alpha = \frac{1 + 1/\sqrt{2}}{2}.$$

Define, for $t, m \ge 0$,

$$\mathcal{F}(t,s) = \sqrt{s(t-s)} + s \quad (0 \le s \le t), \qquad g(t,m) = \mathcal{F}(t, \min(m, \alpha t)). \tag{9.3}$$

Then $\mathfrak{f} \le g(T,m)$. In particular, with $e = P_{01} + P_{10}$, if $e \le 1/4$ then

$$\mathfrak{f} \le \sqrt{e(1-2e)} + e. \tag{9.4}$$

Proof. First fix a transcript and a scalar $\theta > 0$. Normalize the two pulled-back Alice effects as in the source proof; then split at $A_0 \ge \theta A_1$ in a common eigenbasis. For the resulting v, w, set $a = p_{00}(v)$, $d = p_{11}(v)$, $a' = p_{00}(w)$, $d' = p_{11}(w)$. The same two CP block estimates give

$$f_t \le \sqrt{ad} + \sqrt{a'd'} + 2\sqrt{a'd}.$$

Put $s = d + a'$ and $t = a + d + a' + d' = p_{00} + p_{11}$. Cauchy–Schwarz and $2\sqrt{a'd} \le a' + d$ give

$$f_t \le \sqrt{(a + d')(d + a')} + s = \mathcal{F}(t,s).$$

The spectral comparison now yields $d \le p_{01}(v)/\theta$ and $a' \le \theta p_{10}(w)$. Thus $0 \le s \le \min(t, m_t)$ with $m_t = p_{01}/\theta + \theta p_{10}$.

For $t > 0$, differentiating on $0 < s < t$ gives

$$\partial_s \mathcal{F}(t,s) = 1 + \frac{t-2s}{2\sqrt{s(t-s)}}.$$

Its maximum on $[0, \min(m_t, t)]$ is attained at $s_*(t) = \min(m_t, \alpha t)$. At $t = 0$ put $s_* = 0$. Therefore $f_t \le \mathcal{F}(t, s_*)$. This bound uses an explicit Borel function of the observable branch densities: it does not require a measurable choice of the intermediate eigenbasis or spectral split. All terms are integrable since $0 \le s_* \le t$ and $\mathcal{F}(t, s_*) \le 3t/2$.

Choose the same θ for every transcript and set $S = \int s_* \, d\mu$. Scalar Cauchy–Schwarz gives

$$\mathfrak{f} \le \sqrt{\left(\int (t - s_*)\, d\mu\right)\left(\int s_* \, d\mu\right)} + S = \mathcal{F}(T, S),$$

where $S \le T$ and $S \le P_{01}/\theta + \theta P_{10}$. Maximizing once more gives $\mathfrak{f} \le g(T, P_{01}/\theta + \theta P_{10})$. If both disagreement probabilities are positive, choose $\theta = \sqrt{P_{01}/P_{10}}$ to obtain $m = 2\sqrt{P_{01}P_{10}}$. If exactly one is zero, let θ tend to the appropriate zero or infinite limit; continuity of g gives the same bound, including $\mathfrak{f} = 0$. If both vanish, any θ gives it.

Finally $m \le e$ and $T = 1 - e$. The function g is nondecreasing in its second variable. For $e \le 1/4$, one has $e \le \alpha(1-e)$; indeed $T - 2s \ge 1 - 3e > 0$ on $0 \le s \le e$. Hence $g(1-e, e) = \mathcal{F}(1-e, e)$, which is (9.4). □

Corollary 9.2 (Sharpened conditional secret-bit gap)**.** *Under the same represented-class, admitted-protocol, completion, purification, and complete-record hypotheses as the source's secret-bit theorem, every ideal-bit comparison has unhalved error*

$$\eta \ge r_* := \frac{3 - \sqrt{3}}{6} \simeq 0.2113248654. \tag{9.5}$$

The conventional trace-distance bound is $r_*/2 \simeq 0.1056624327$.

Proof. Assume $\eta < r_*$. The unchanged source bookkeeping gives $e \le \eta/2 < 1/4$ and $\mathfrak{f} \ge (1-\eta)/2$. The function $h(e) = \sqrt{e(1-2e)} + e$ is increasing on $[0, 1/4]$: its interior derivative is $1 + (1 - 4e)/(2\sqrt{e(1-2e)}) > 0$. Theorem 9.1 therefore gives

$$\frac{1-\eta}{2} \le \sqrt{\frac{\eta}{2}(1-\eta)} + \frac{\eta}{2}, \qquad \frac{1-2\eta}{2} \le \sqrt{\frac{\eta(1-\eta)}{2}}.$$

Both sides of the second inequality are nonnegative in this range. Squaring yields $6\eta^2 - 6\eta + 1 \le 0$, impossible below its smaller root r_*. Also $r_* > 1/5$, since $\sqrt{3} < 9/5$. □

The optimized imbalance bound is strictly more informative than retaining only the total disagreement probability. Nevertheless no optimal constant is claimed. Tensor closure and the unchanged transcript/recovery proofs are still required at each copy number. The source's public-conditioning, vanishing-abort, and recoverable completed-output-limit arguments carry this new constant through by the same homogeneity and contraction arguments; their hypotheses are not enlarged. Exact finite checks of the threshold arithmetic are diagnostic evidence, not substitutes for the universal quantified theorem just proved.

9.6 Perturbation does not prove robust zero key

For fixed n, suppose a reference state ω has a proved gap g for the chosen completed protocol class, and a perturbed state has purifications $|\psi\rangle, |\phi\rangle$ with $\||\psi\rangle\langle\psi| - |\phi\rangle\langle\phi|\|_1 \le \delta$. The tensor telescoping identity and channel contraction give a fixed-n output comparison at most $n\delta$, hence a gap at least $g - n\delta$ for the same normalized completed protocol on both inputs. This bound is not uniform in n. In particular, our robust entanglement witness does not imply a neighborhood of zero distillable key. Completion on only one input also does not justify this channel comparison without a normalized completion on the other.

10 Other Source Constructions and Public Comparisons

10.1 The flag channel and the independent large construction

For a PPT pair $\Psi_i : \mathrm{M}_m \to \mathrm{M}_m$, the source sets $A_i = \Psi_i^\dagger(I)$, $c_i = (1 + \operatorname{tr} A_i)^{-1}$, $F_i = I - c_i A_i \succeq c_i I$. On $\mathbb{C}_0^m \oplus \mathbb{C}_1^m \oplus \mathbb{C}e_\star$, the channel switches the scaled maps between the two blocks, discards cross blocks, and sends missing trace to an absorbing flag. Every branch is CP, the trace identity is $c_i A_i + F_i = I$, and the output transpose is CP because the moving branches are PPT and the flag is real. Squaring gives the local corner $c_1 c_2 J(\Psi_2 \Psi_1)$. Local compression preserves separability, so nonseparability of that corner suffices. This is an inherited construction, not a new contribution of our report.

The alternative projection criterion uses real unit vectors $u_q \in \mathbb{R}^K$ and projective measurements P_i on $\mathbb{C}^D$, with $H_{ij} = (u_q^T u_r) P_i P_j$. A block Gram factorization proves positivity; commutation whenever $u_q^T u_r \neq 0$ proves PPT. For $C = J(\Psi^\dagger \Psi)$ one has $C_{(i,j),(i,j)} = \operatorname{tr}(P_i P_j)$ and the diagonal-subspace compression $B_{ij} = (u_q^T u_r)^2 \operatorname{tr}(P_i P_j)$. Its entry sum is at least $(Q/K) \operatorname{tr} B$. A separable decomposition would force each compressed vector to have support of size at most κ, yielding an upper bound $\kappa \operatorname{tr} B$. The strict inequality $\kappa < Q/K$ is indispensable.

The source realizes it with $p = 257$, $k = 1028$, $b = \binom{1027}{513}$, $a_0 = \sum_{j=0}^{256} \binom{1028}{j}$, $Q = b^3$, $K = k^3 = 1\,086\,373\,952$, and $\kappa = 3a_0 b^2$. Restricted-intersection polynomials have degree at most $p - 1$, giving the bound a_0; fiber induction gives eight orthogonal tuples in sufficiently large subsets. Six Mermin–Peres contexts on each orthogonal eight-subset impose an inconsistent parity assignment on any all-positive-overlap choice. The tensor dimension is $D = 4^{\binom{Q}{8}}$ and the label count $m = QD$. The proof uses a finite formula at an enormous dimension, not a matrix that has been computed here. Our exact arithmetic confirms $2^{-194} \le a_0/b < 2^{-193}$ and $3a_0 K < b$. The 4096-selection parity enumeration and its five-bin histogram in the appendix were read but not rerun. The same local channel on two halves is distinct from the square of an endomap, and neither formulation removes the independent operational dependency.

10.2 Current primary public literature

Public checks on October 8, 2026 used public mathematical topics and titles only. The primary private-state work of Horodecki and coauthors establishes that bound entanglement can yield secret key [2, 3]; consequently "PPT, therefore zero key" is not a valid shortcut. The LOCC formalism of Chitambar and coauthors distinguishes finite and unbounded-round protocols and emphasizes the nonclosed nature of LOCC [4]. This supports treating completion and closure as explicit assumptions, not interchangeable labels.

Christandl, Müller-Hermes, and Wolf prove positive PPT-composition results in dimension three and for Gaussian channels, and formulate related equivalent conjectures [5]. Their Choi and

composition machinery is classical prior work for this audit. The July 2026 preprint of An and Lee proves a qutrit statement for one-copy-undistillable maps composed with maps of Schmidt number at most two [6]; its dimension and hypotheses do not settle the proposed dimension-ten construction. Park's August 2026 preprint states eventual entanglement-breaking behavior for every PPT linear map [7]. An eventual iterate bound, even if established, need not be two and is logically compatible with a non-entanglement-breaking square. These recent preprint claims are reported at their stated scope, not adopted as independently verified theorems here.

The source's qualitative product exclusion is related to unextendible product bases [8]. Our witness uses that classical separation principle, a full-spark frame lower bound, and a new-to-this-report residual accounting for the specific exterior-square construction. We claim no historical priority for these general ingredients.

11 Verification, Limitations, and Research Priorities

11.1 Executed versus unexecuted work

The bounded checker is `_audit/M553-author-check.py`; its output is `_audit/M553-author-check-results.json`. It reads the displayed matrices directly from the source, asserts their shapes, checks every integer block equality, reconstructs fifteen determinant polynomials, replays three modular characteristic polynomials and sixty gcd degrees, checks modular border commutators, checks the Choi partial transpose and rank, computes (6.4), and checks the stated small combinatorial inequalities. It additionally checks the exterior-dual determinant and squared-minor identities on three exact Gaussian-integer fixtures, including two non-real vectors. All arithmetic uses Python standard-library integers and exact fractions. These three fixtures check implementation conventions; the universal identity is proved in Theorem 7.1.

A separate exact checker in `_audit/M553-independent-20261008/exact_audit.py` also reproduces the modular data using characteristic-polynomial interpolation rather than Newton identities. Its recorded rational computation gives

$$\det H = 464783813657532325475350044847240672350021206346150445056 0$$

and zero rational commutators for all three coordinate multiplication operators. It also reproduces the 4096-choice parity histogram. These are independently recorded checks, not additional executions of the author checker. The accompanying written audit found no blocking defect in the source arguments it examined; this is targeted internal review, not formal verification or external peer review.

The program does not run a general Galois-group solver, isolate twenty algebraic roots, enumerate all ten-point determinants, assemble Z in full, instantiate the alternative huge matrices, simulate protocols, or compute a distillable-key rate. The exact frame margin s and any numerical robustness radius remain uncomputed. A full source reading and matched modular data do not constitute external peer review.

The local results fully proved here are the realness criterion and its trace-metric consequence, the determinant/minor residual formula, the full-spark witness and distance bound, and the error-budget and physical moment-coordinate transfers, together with the retained-mass fidelity refinement and sharpened conditional gap. Some are conditional statements with explicit premises, particularly a positive frame margin. The exact normalization is realized computationally. These are the evidence for C3, not the inherited headline theorem, report length, applicability L5, or any claimed speedup.

11.2 Pros, costs, and next tests

The principal benefit is a clean separation of identity checks, projective independence, robust finite entanglement, and operational secrecy. The squared-minor formula turns an algebraic near-zero into a quantity with an exact meaning for the witness. It also bypasses an unjustified projection of an approximate root onto the exact ideal. The six-dimensional normalization avoids constructing 100×100 entries merely to find their trace, but no timed comparison is claimed. The moment bank retains a known physical metric, so a coordinate change cannot manufacture a false frame margin.

The costs are real: the subset minimum may be ill-conditioned, algebraic heights may be large, and raw monomial moments can be poorly conditioned. A Galois proof of nonvanishing is not a usable lower bound for numerical determinants. Conversely a good numerical condition number is not proof of all-subset independence. Stronger quantitative constants would be secondary to obtaining even one complete certified margin for the actual source state.

Priority next work is exact real-root isolation with certified Veronese normalization, lower frame bounds for all subsets or a rigorous symmetry reduction, and an independent implementation of the composite-trace formula. A separate operational review should focus on common local factors for terminal readouts, the treatment of infinite histories, and preservation of Eve's complete information under limits. One must not extend a fixed-copy perturbation estimate to all copies without a new uniform argument. These local contributions do not independently validate the headline construction and all-protocol theorem.

A Reproduction and Read-Scope Record

The bundled interpreter used is

> `C:/Users/mural/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/python.exe`.

Run the author checker with assertions enabled and without Python optimization. Its loops are bounded: four source matrices, fifteen minors, three primes, twenty characteristic steps per prime, twenty Frobenius powers per prime, and finite small-rank elimination. The recorded input SHA-256 values include every available local source file, the catalogue at read time, the contribution policy, and all three private-corpus files. The catalogue hash identifies the read-time input; subsequent catalogue revisions are not part of that evidence snapshot.

The exact substantive read scope was `sections/introduction.tex`, `sections/foundations.tex`, `sections/coherence.tex`, `sections/transcripts.tex`, `sections/secret-key.tex`, `sections/geometry.tex`, `sections/certificate.tex`, `sections/applications.tex`, `sections/projection.tex`, `sections/combinatorial.tex`, `appendices/arithmetic.tex`, and `appendices/supplementary-checks.tex`. All were read in full; transcript continuation was separately read to avoid output truncation. Private-corpus scope is exactly as listed in Section 8, not the entire multi-volume collection. Routing maps were not used as substitutes for those passages.

References

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